

Solutions to Homework #3

Problem 1

By Euler's identity, we have

$$x[n] = \frac{1}{2}e^{j\pi n/2} + \frac{1}{2}e^{-j\pi n/2}.$$

Then

$$y[n] = h[n] * x[n] = \frac{1}{2}H(e^{j\pi/2})e^{j\pi n/2} + \frac{1}{2}H(e^{-j\pi/2})e^{-j\pi n/2},$$

where

$$H(e^{j\pi/2}) = e^{-j\pi/4} \cdot \frac{1 + e^{-j\pi} + 4e^{-j2\pi}}{1 + \frac{1}{2}e^{-j\pi}} = 8e^{-j\pi/4}$$

and

$$H(e^{-j\pi/2}) = e^{j3\pi/4} \cdot \frac{1 + e^{j\pi} + 4e^{j2\pi}}{1 + \frac{1}{2}e^{j\pi}} = 8e^{j3\pi/4}.$$

Combining the above results, we obtain

$$y[n] = 4e^{j(\pi n/2 - \pi/4)} + 4e^{j(-\pi n/2 + 3\pi/4)}.$$

The average of the exponent is

$$\frac{1}{2} \left(\frac{\pi n}{2} - \frac{\pi}{4} - \frac{\pi n}{2} + \frac{3\pi}{4} \right) = \frac{\pi}{4}.$$

Therefore, $y[n]$ is rewritten as

$$y[n] = 4e^{-j\pi/4} \left(e^{j(\pi n/2 - \pi/2)} + 8e^{j(-\pi n/2 + \pi/2)} \right).$$

By using Euler's identity, the expression of $y[n]$ simplifies to

$$y[n] = 8e^{-j\pi/4} \cos \left(\frac{\pi n}{2} - \frac{\pi}{2} \right) = 8e^{-j\pi/4} \sin \left(\frac{\pi n}{2} \right).$$

There was an error in the previous version of the solution, which assumed that the fact that the output of the system $H(e^{j\omega})$ to the input $\cos(\omega_0 n)$ is

$$y[n] = |H(e^{j\omega_0})| \cos(\omega_0 n + \angle H(e^{j\omega})). \quad (1)$$

However, this expression is valid only when the frequency response $H(e^{j\omega})$ satisfies the conjugate-symmetry property

$$H(e^{-j\omega}) = H^*(e^{j\omega}),$$

which holds if and only if $h[n]$ is real-valued.

In Problem 1, the frequency response $H(e^{j\omega})$ does not satisfy this conjugate symmetry. Therefore, applying (1) and obtaining

$$y[n] = 8 \cos\left(\frac{\pi n}{2} - \frac{\pi}{4}\right).$$

is not correct.

Problem 2

Since $\{x[n]\}$ is periodic, we have

$$x[n + N] = x[n], \quad \forall n \in \mathbb{Z}.$$

The output is given by

$$\{y[n]\} = \{h[n]\} * \{x[n]\}.$$

In other words,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n - k], \quad \forall n \in \mathbb{Z}$$

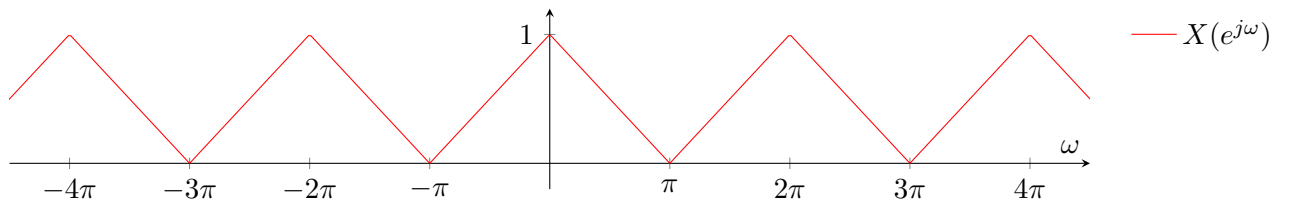
Then, for any n , we have

$$\begin{aligned} y[n + N] &= \sum_{k=-\infty}^{\infty} h[k]x[(n + N) - k] \\ &= \sum_{k=-\infty}^{\infty} h[k]x[(n - k) + N] \\ &= \sum_{k=-\infty}^{\infty} h[k]x[n - k] \\ &= y[n], \end{aligned}$$

where the third step follows from the periodicity of $\{x[n]\}$. Therefore, we claim that $\{y[n]\}$ is also periodic with period N .

Problem 3

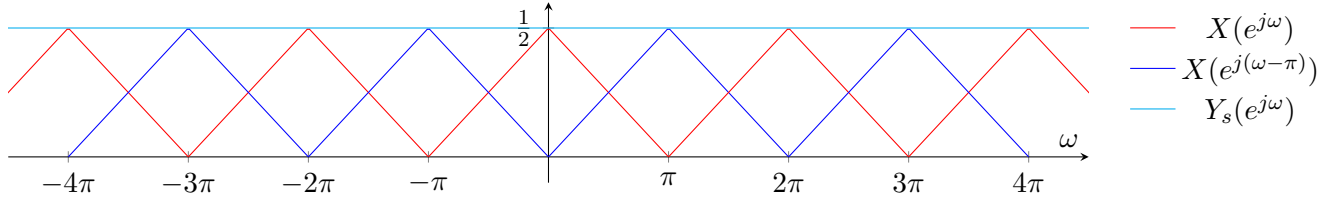
By the definition, $X(e^{j\omega})$ is 2π -periodic. The given signal is plotted as follows.



- (a) Note that $(-1)^n$ is rewritten as $e^{j\pi n}$. Therefore, $(-1)^n x[n] = e^{j\pi n} x[n]$ is considered as the modulation in time, which corresponds to $X(e^{j(\omega-\pi)})$. Hence, we obtain

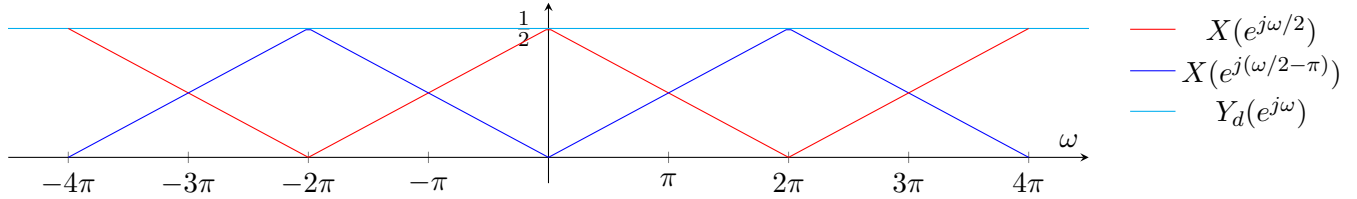
$$Y_s(e^{j\omega}) = \frac{1}{2}X(e^{j\omega}) + \frac{1}{2}X(e^{j(\omega-\pi)}),$$

which is illustrated below.



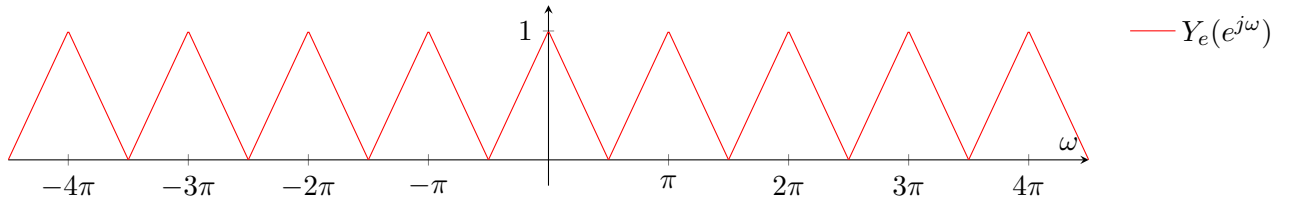
- (b) Since $y_d[n] = x[2n]$, we have

$$Y_d(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[2n]e^{-j\omega n} = \sum_{m=-\infty}^{\infty} y_s[m]e^{-j\frac{\omega}{2}m} = Y_s(e^{j\omega/2})$$



- (c) The DTFT of $y_e[n]$ is written as

$$Y_e(e^{j\omega}) = \sum_{m=-\infty}^{\infty} x[m]e^{-j2\omega m} = X(e^{j2\omega}).$$



Problem 4

- (b) Since $f(\omega)$ is an impulse train where the consecutive impulses are separated by $\frac{2\pi}{L}$, we can deduce that it is T periodic for $T = \frac{2\pi}{L}$. The Fourier series coefficient is computed as

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} f(\omega) e^{-j2\pi k\omega/T} d\omega \stackrel{i)}{=} \frac{L}{2\pi} \int_{-\pi/L}^{\pi/L} \delta(\omega) e^{-jLk\omega} d\omega \stackrel{ii)}{=} \frac{L}{2\pi},$$

where i) holds since we have a single impulse on the given interval for the integration and ii) follows from the sifting property of the Dirac delta. Note that we wanted to compute the integral on $[-T/2, T/2)$ instead of $[0, T)$ to avoid the pulse on the boundary.

- (d) By the construction of $x[n]$, we have

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \frac{L}{2\pi} \sum_{k=-\infty}^{\infty} e^{j\omega Lk} = \sum_{k=-\infty}^{\infty} a_k e^{j\omega Lk} = f(\omega).$$

Therefore, we deduce that

$$\mathcal{F}_{\text{DTFT}} \left\{ \sum_{m=-\infty}^{\infty} \delta[n + Lm] \right\} = \frac{2\pi}{L} \sum_{l=-\infty}^{\infty} \delta \left(\omega - \frac{2\pi l}{L} \right).$$

- (e) By the convolution theorem, we have

$$Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} H(e^{j\pi l/8}) \delta \left(\omega - \frac{\pi l}{8} \right).$$

It suffices to observe $Y(e^{j\omega})$ on the interval $[-\pi, \pi)$. Since $H(e^{j\omega})$ is a lowpass filter, we have

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{l=-1}^1 H(e^{j\pi l/8}) \delta \left(\omega - \frac{\pi l}{8} \right) \\ &= e^{j3\pi/8} \delta(\omega + \pi/8) + \delta(\omega) + e^{-j3\pi/8} \delta(\omega - \pi/8). \end{aligned}$$

Then

$$y[n] = \frac{1}{2\pi} e^{-j\pi n/8 + j3\pi/8} + \frac{1}{2\pi} + \frac{1}{2\pi} e^{j\pi n/8 - j3\pi/8} = \frac{1}{2\pi} + \frac{1}{\pi} \cos \left(\frac{\pi(n-3)}{8} \right).$$

Problem 5

(a) First, recall that z -transform for any signal $x[n]$ can be written as

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}.$$

Therefore, given $x[n] = 0$ if $n < 0$ or $n \geq N$, $X(z)$ reduces to

$$X(z) = \sum_{n=0}^{N-1} x[n]z^{-n} = \sum_{n=0}^{N-1} x[n] (z^{-1})^n.$$

Thus, $X(z)$ is a polynomial in z^{-1} with degrees $N - 1$.

Next, note that $h[n] \neq 0$ for $n = 0, 1, 2$ only. In the z -domain, we have

$$H_k(z) = \sum_{n=0}^2 h_k[n]z^{-n} = \sum_{n=0}^2 h_k[n] (z^{-1})^n, \quad k = 1, 2.$$

Therefore, $H_k(z)$ is a polynomial in z^{-1} with degrees 2 for $k = 1, 2$.

Finally, it is given that $y_k[n] = h_k[n] * x[n]$ for $k = 1, 2$. In the z -domain, convolution becomes multiplication, i.e.

$$Y_k(z) = H_k(z)X(z).$$

Using result from part (a) and (b),

$$\begin{aligned} Y_k(z) &= \sum_{l=0}^2 h_k[l] (z^{-1})^l \sum_{n=0}^{N-1} x[n] (z^{-1})^n \\ &= (h_k[0] + h_k[1]z^{-1} + h_k[2]z^{-2}) \sum_{n=0}^{N-1} x[n] (z^{-1})^n \\ &= \sum_{n=0}^{N-1} \left(h_k[0]x[n] (z^{-1})^n + h_k[1]x[n] (z^{-1})^{n+1} + h_k[2]x[n] (z^{-1})^{n+2} \right). \end{aligned}$$

Thus, $Y_k(z)$ is a polynomial in z^{-1} with degrees $N + 1$.

(b) Given $\hat{X}(z) = \alpha X(z)$, $\hat{H}_1(z) = \frac{1}{\alpha} H_1(z)$, and $\hat{H}_2(z) = \frac{1}{\alpha} H_2(z)$, we have

$$\hat{Y}_k(z) = \hat{H}_k(z)\hat{X}(z) = \frac{1}{\alpha} H_k(z)\alpha X(z) = H_k(z)X(z), \quad k = 1, 2$$

Since $\hat{Y}_k(z) = Y_k(z)$, we can not distinguish between $(\hat{y}_1[n], \hat{y}_2[n])$ from $(y_1[n], y_2[n])$. Therefore, we can only obtain the signal parameters $x[n], h_1[n], h_2[n]$ up to scaling ambiguity.

- (c) Using the fundamental theorem of algebra, the polynomials $H_1(z)$ and $H_2(z)$ are factorized as

$$H_k(z) = h_k[2] \prod_{j=0}^2 (z^{-1} - \gamma_{k,j}), \quad k = 1, 2.$$

Therefore, we obtain

$$\frac{Y_2(z)}{Y_1(z)} = \frac{H_2(z)X(z)}{H_1(z)X(z)} = \frac{H_2(z)}{H_1(z)} = \frac{h_2[2] \prod_{j=0}^2 (z^{-1} - \gamma_{2,j})}{h_1[2] \prod_{j=0}^2 (z^{-1} - \gamma_{1,j})}.$$

Thus if $H_1(z)$ and $H_2(z)$ do not share any common root, the roots of $H_1(z)$ and $H_2(z)$ can be uniquely determined respectively from the roots of the denominator and numerator of $Y_2(z)/Y_1(z)$. Once we determine the roots of $H_1(z)$ and $H_2(z)$, the roots of $X(z)$ is determined from the other roots of $Y_1(z)$ and $Y_2(z)$.